

Heritage Treasures: An In-Depth Analysis of
UNESCO World Heritage Sites in Tableau

TEAM ID	SWTID-2026-4638
PROJECT TITLE	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
MAX MARKS	3 MARKS

3.4 – Technology Stack

The realization of the "Heritage Treasures" analytical platform relies on a modern, highly optimized technology stack tailored for sophisticated data engineering, spatial analysis, and visual business intelligence. The core of the analytics workspace is Tableau Desktop 2025.3.1, which provides the advanced visualization engine, spatial mapping tools, and parameter controls needed to build the interactive dashboards. This version introduces enhanced performance optimizations for handling complex coordinates and multi-layered data models, ensuring smooth interactions even when rendering thousands of individual global heritage sites simultaneously. The data storage and processing layer utilizes the native Tableau Hyper data engine technology. Hyper is an in-memory database system designed for fast analytical query processing on large, complex datasets. By converting raw CSV and Excel data sources into a optimized `.hyper` extract, the project benefits from rapid column-store performance. This setup allows complex level-of-detail (LOD) expressions, cross-database joins, and intensive spatial calculations to execute almost instantly, avoiding the performance lags often found in traditional relational databases. For data preparation, validation, and structuring, the environment uses a combination of Tableau Prep Builder and advanced text processing tools to normalize the raw UNESCO source files. The data schema is built on a star-like logical relationship layout, handling multi-language strings, geographic locations.